

Information Technologies for Industrial Engineers

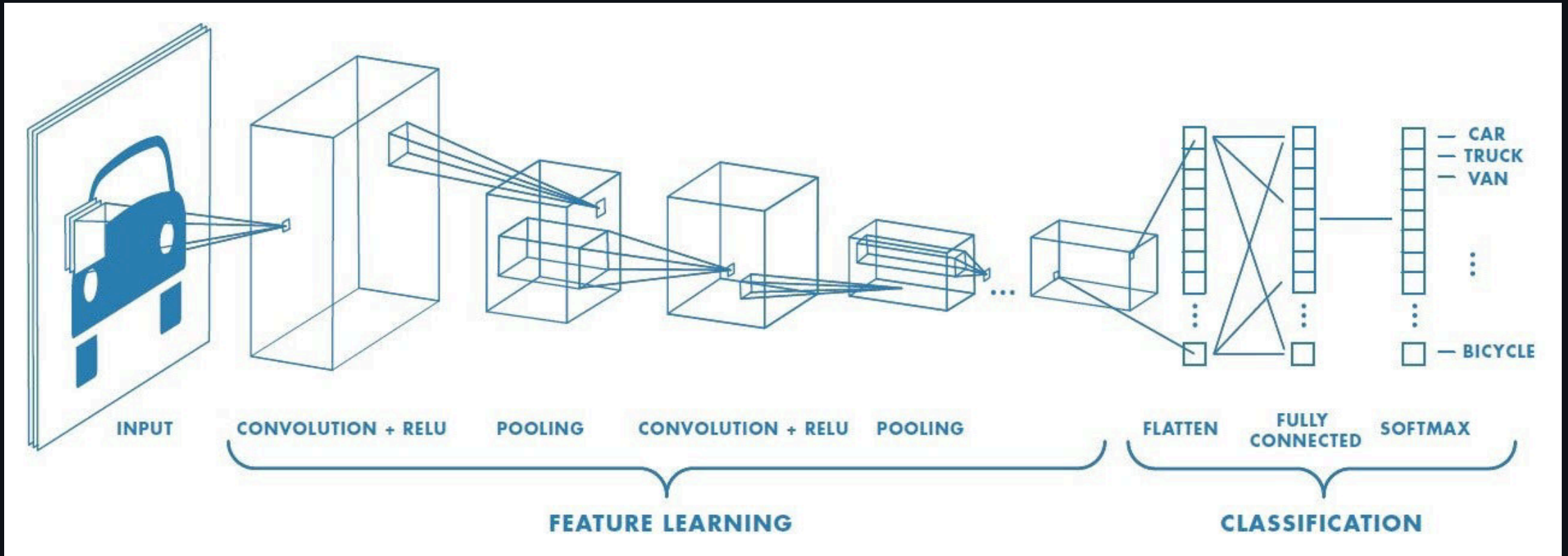
เทคโนโลยีสารสนเทศสำหรับวิศวกรอุตสาหกรรม

Computer vision

- Field of computer science that focuses on enabling computers to identify and understand objects and people in images and videos.
- Computer vision seeks to perform and automate tasks that replicate human capabilities.
 - The way humans see.

Convolutional neural network

- Deep Learning algorithm that can take in an input image and be able to differentiate one from the other.
- The architecture of a **ConvNet** is analogous to that of the connectivity pattern of Neurons in the human brain and the visual cortex.



Source

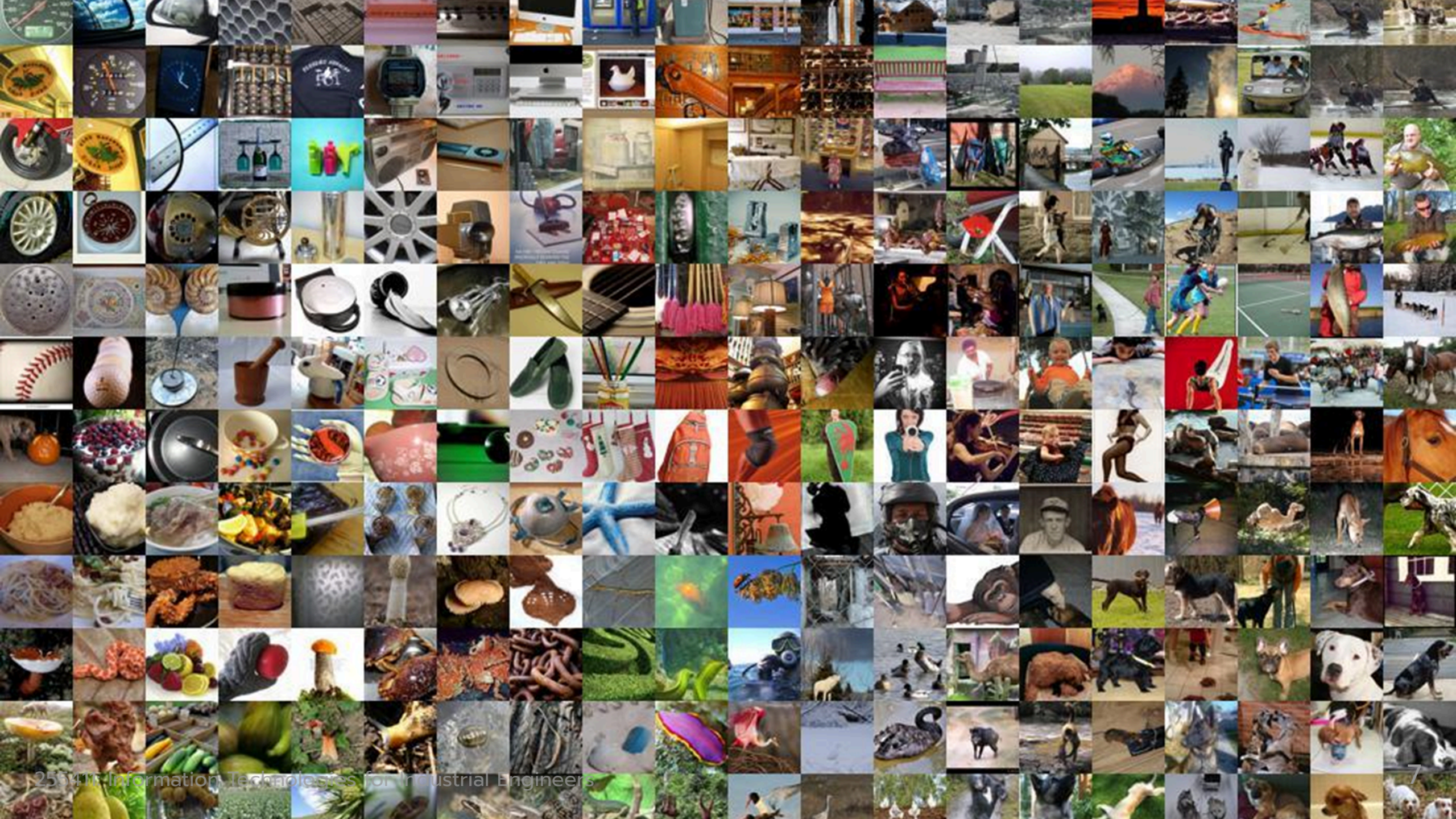
Image classification

- Dog (50%)
- ... (15%)
- ... (15%)

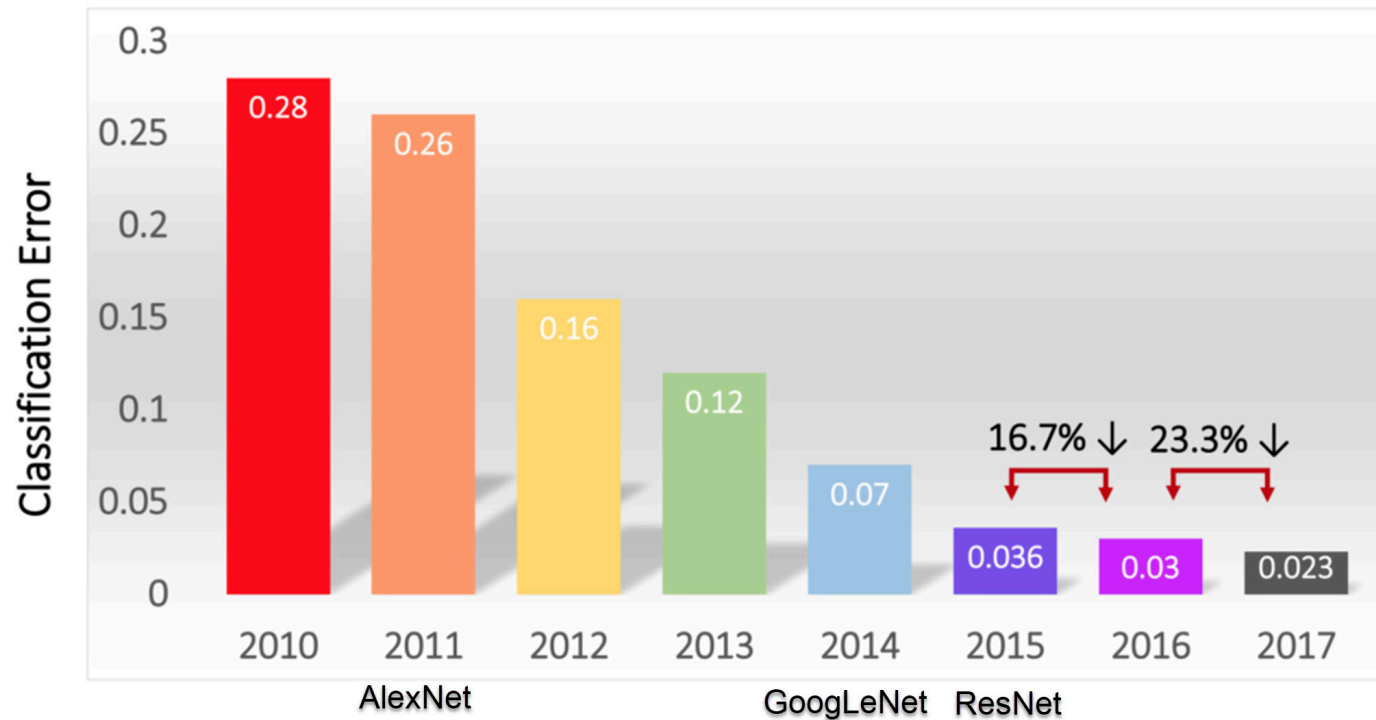


ILSVRC competition

- ImageNet Large Scale Visual Recognition Challenge
 - Started 2010
- Task
 - Evaluates algorithms for object detection and image classification at large scale.
- Data
 - 15 million images
 - 22000 categories



Classification Results (CLS)



- Y-axis: Top-5 error
- Exceed human ability @ 2015

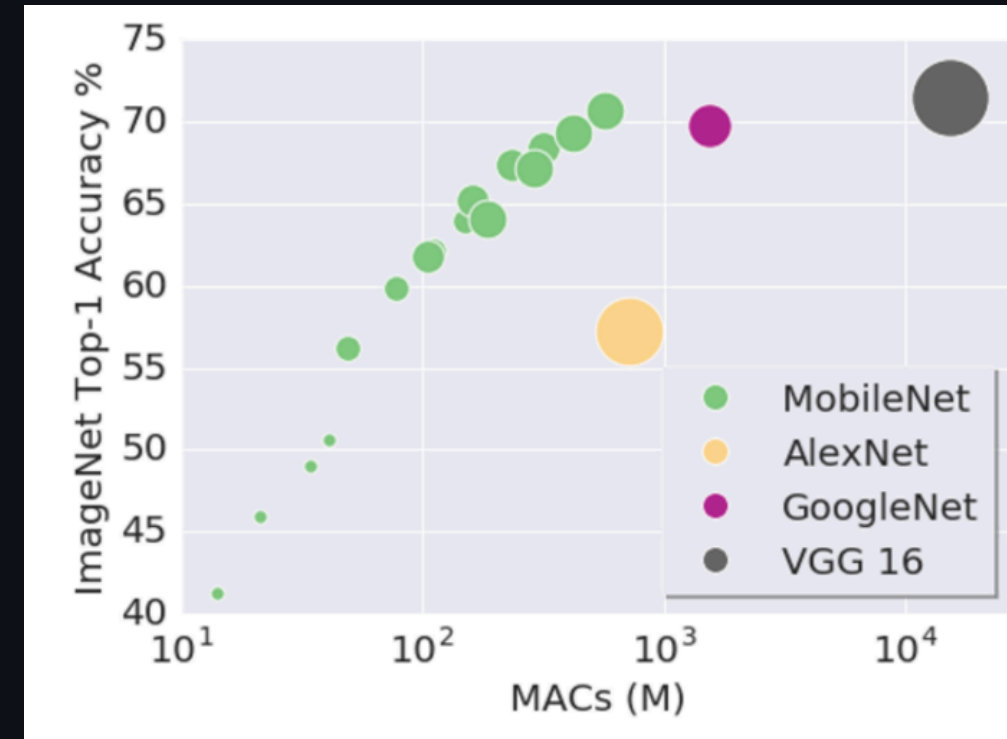
Current ranking

<https://paperswithcode.com/sota/image-classification-on-imagenet>

| (Link is not working, unfortunately. See [this](#) instead)

MobileNet

- MobileNet is a simple but efficient and not very computationally intensive convolutional neural networks for mobile vision applications
- [Info](#)



Model File

- ONNX

qualcomm/**MobileNet-v3-Large** like 1 Follow

Image Classification PyTorch LiteRT backbone real_time ai

Model card Files and versions Community

v0.32.0 MobileNet-v3-Large

qaihm-bot v0.32.0 d6c4f7b VERIFIED

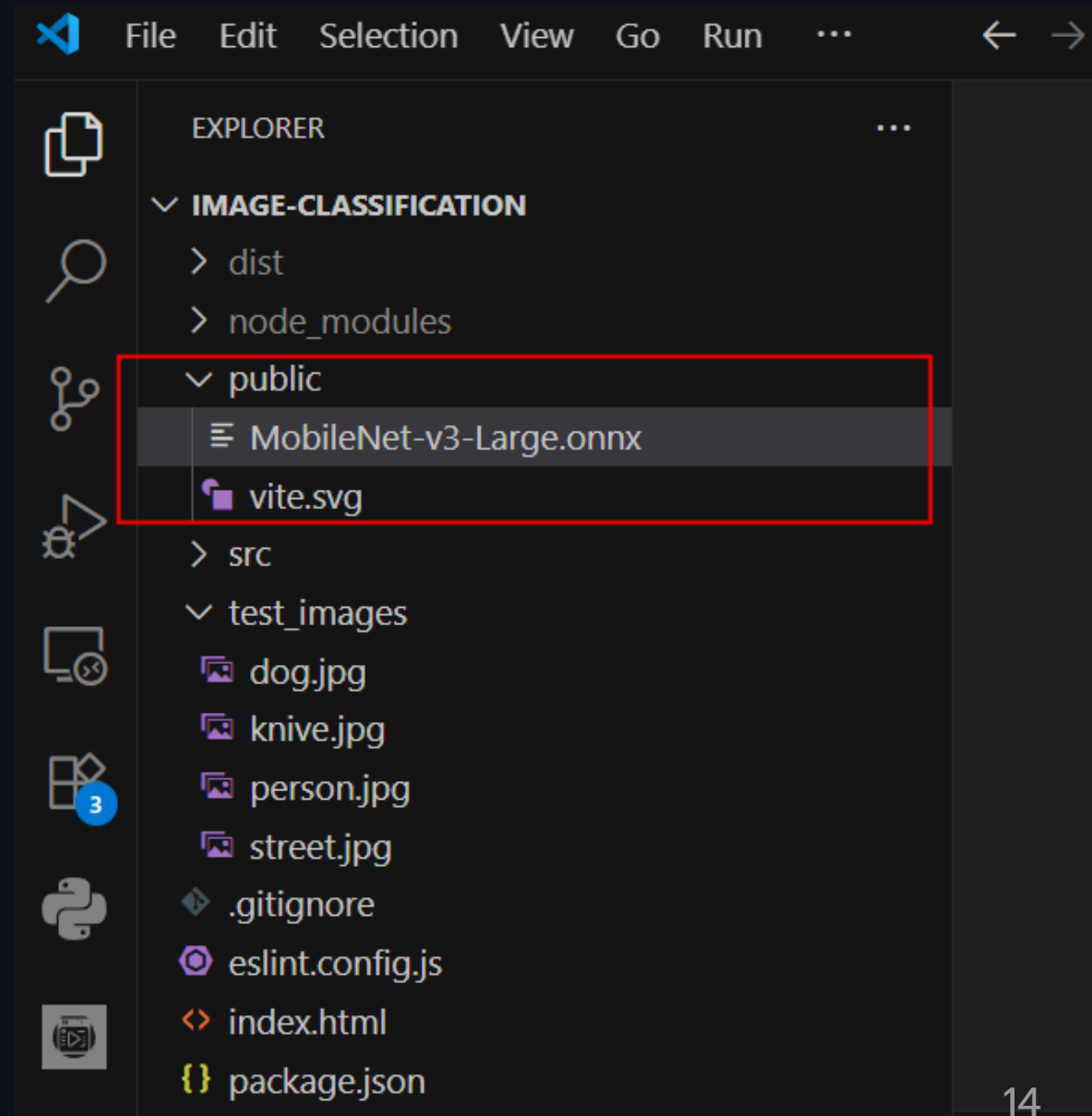
- .gitattributes Safe
- DEPLOYMENT_MODEL_LICENSE.pdf Safe
- LICENSE Safe
- MobileNet-v3-Large.dlc Safe
- MobileNet-v3-Large.onnx** Safe
- MobileNet-v3-Large.tflite Safe
- MobileNet-v3-Large_w8a16.dlc Safe
- MobileNet-v3-Large_w8a16.onnx Safe
- README.md Safe

Let's do it

Install

- `pnpm create vite@latest`
- ...
- `pnpm i onnxruntime-web jimp`
- `pnpm approve-builds`
 - Approve all packages.
- `pnpm i -D @jimp/types`

Image Model



Setup

- `./vite/config.ts`
 - Make `onnx` extension available in `public` folder.
- `./main.tsx`
 - Remove `css` import.
- `./index.html`
 - Add `PicoCSS`

Core Code

- `./src/model.ts`
- `./src/ml.ts`
- `./src/classes.ts`
- `./src/App.tsx`

Build and Deploy

- `pnpm run build`
- Deploy on `Netlify`

Limitation

There are many things in this image.



Predictions

School Bus (100% likely)

Trolleybus (0% likely)

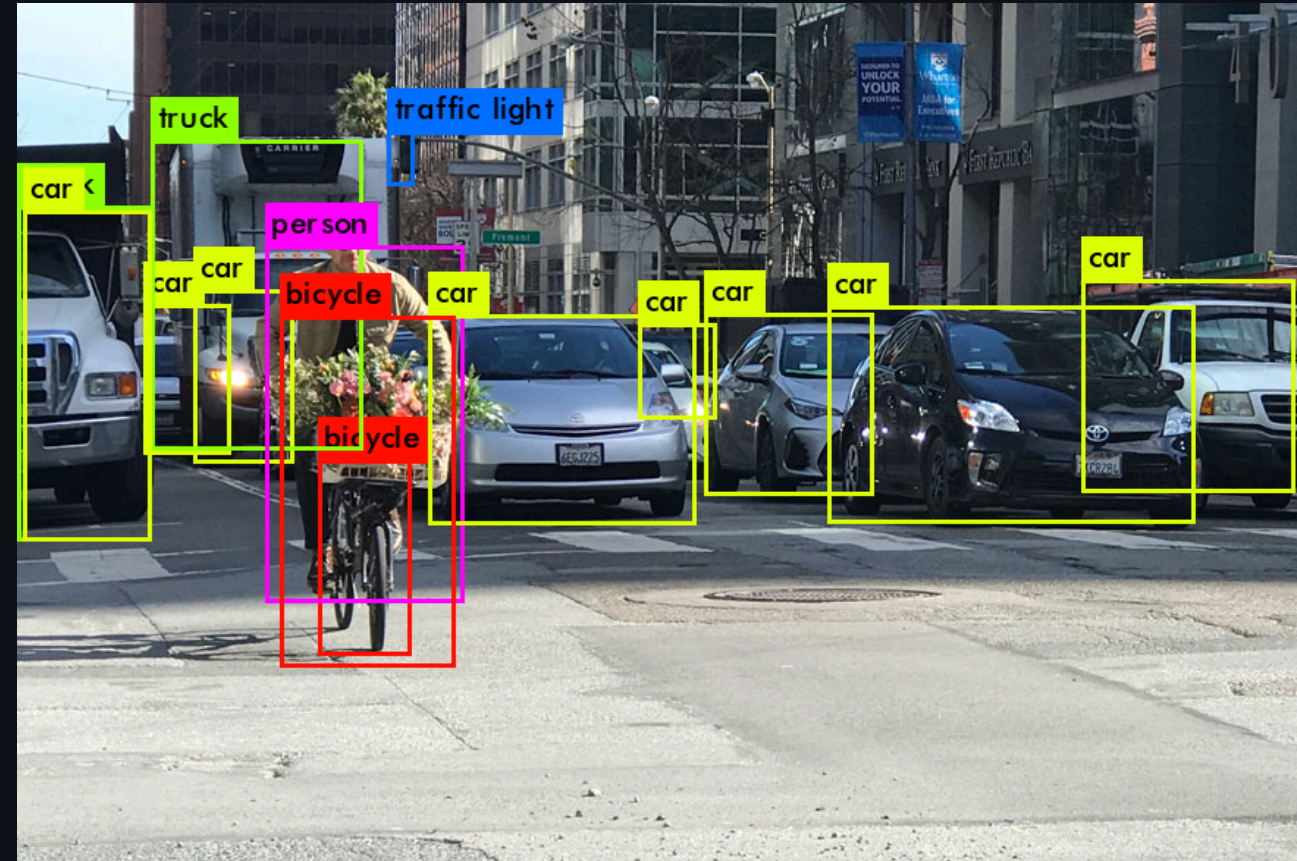
Passenger Car (0% likely)

Minibus (0% likely)

Recreational Vehicle (0% likely)

Object detection

- Car
 - Top: 500, Bottom: 200, Left: 50, right: 400
 - 50%
- Bicycle
 - ...
 - ...



Models

- **YOLO** (You Only Look Once)
 - Grid-based approach
 - Faster, less accurate
- **SSD** (Single Shot Detector)
 - Feature-map approach
 - Faster, less accurate (comparable to YOLO)
- **R-CNN** (Region-based Convolutional Neural Network)
 - Pixel classification
 - Slower, more accurate

Source

COCO dataset

- *Common Objects in Context*
- Large-scale image recognition dataset for object detection, segmentation, and captioning tasks.
 - Contains over 330,000 images.
 - Annotated with 80 object categories.
- <https://cocodataset.org/#explore>

SSD-MobilenetV1

- This model detects objects defined in the COCO dataset.
- Uses SSD algorithm
- Classify using MobilenetV1.

Model Download

[Link](#)

Model

Model	Download	Download (with sample test data)	
SSD-MobilenetV1	29.3 MB	27.9 MB	
SSD-MobilenetV1-12	28.1 MB	24.3 MB	
SSD-MobilenetV1-12-int8	9.2 MB	5.6 MB	
SSD-MobilenetV1-12-qdq	9.6 MB	5.7 MB	

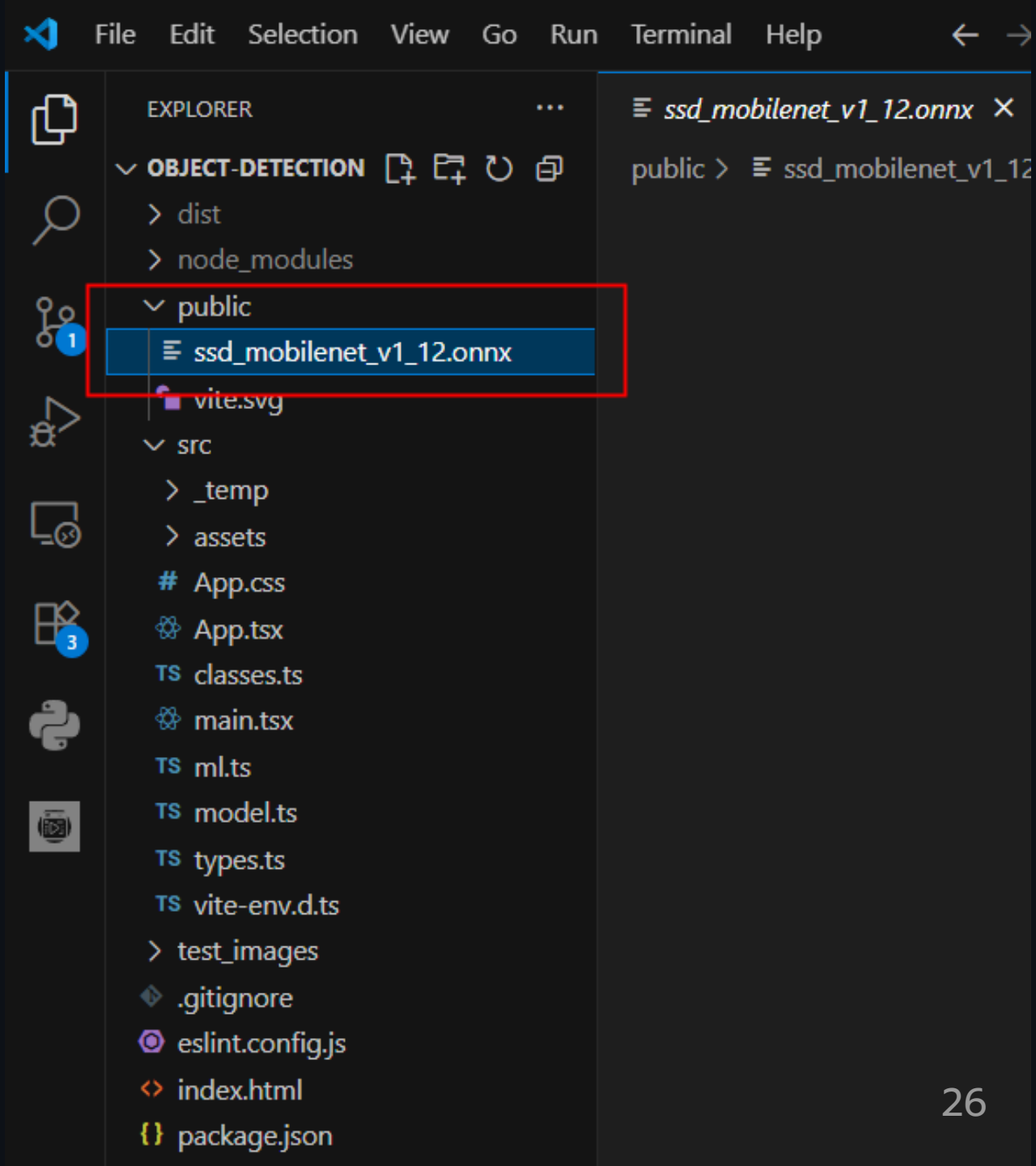
Compared with the fp32 SSD-MobilenetV1-12, int8 SSD-MobilenetV1-12's mAP dr
1.23x.

Object Detection App

Setting up

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- `pnpm i -D @jimp/types`

Model



Setup

- `./vite/config.ts`
 - Make `onnx` extension available in `public` folder.
- `./main.tsx`
 - Remove css import.
- `./index.html`
 - Add `PicoCSS`
- `./src/App.css`
 - Custom CSS for display

Core Code

- `./src/model.ts`
- `./types.ts`
- `./src/ml.ts`
- `./src/classes.ts`
- `./src/App.tsx`

Build and Deploy

- `pnpm run build`
- Deploy on `Netlify`

Object Detection with Webcam

<https://it-for-ie-test-coco.netlify.app>

- Source Code
- Using Tensorflow library